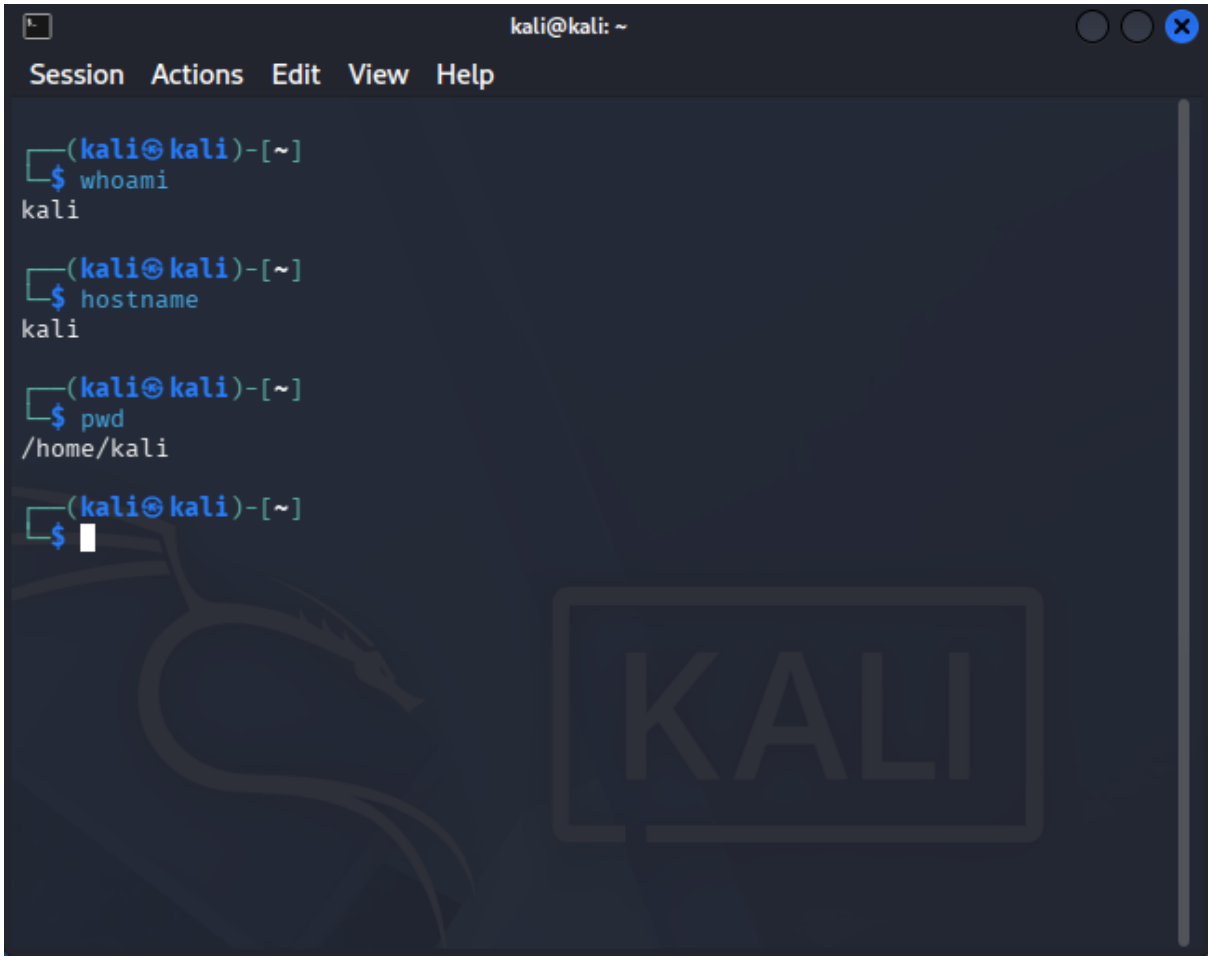
OMAMUYOVWI GBOBOH

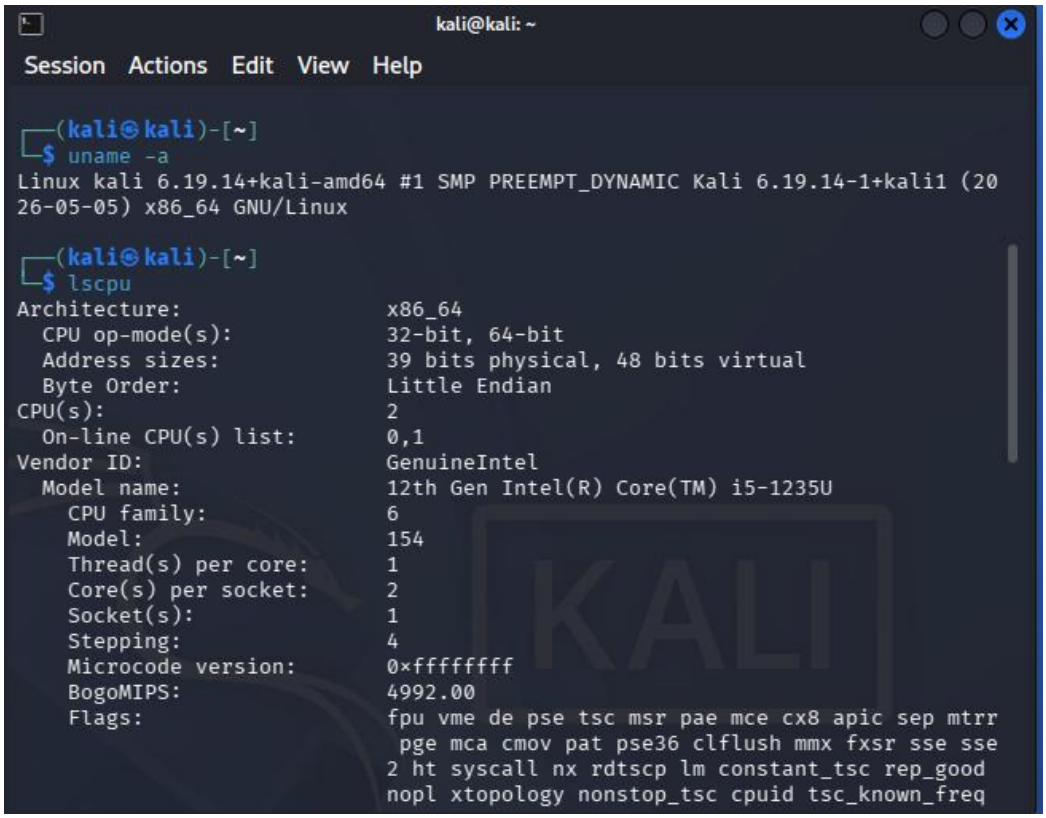
GRC-C26-09-EU-035

24th September 2026

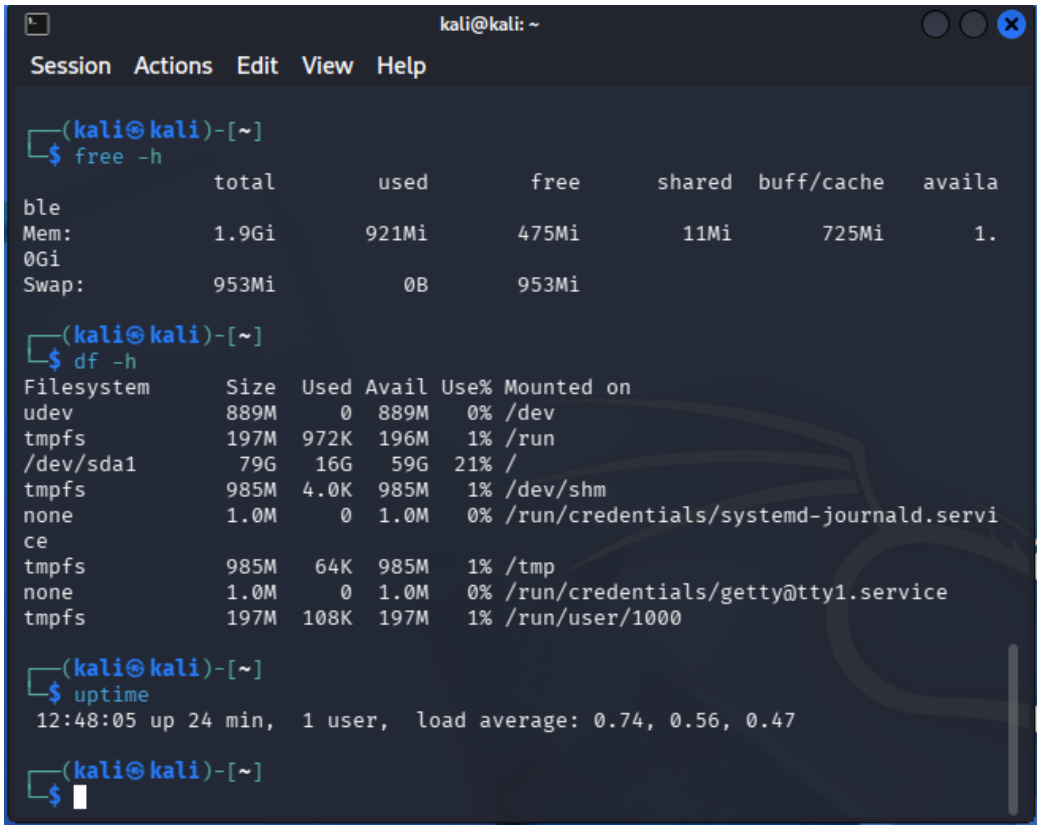


1. My username is **Kali**, the machine hostname is **Kali,** and I am in the **home directory**.

2.A


kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ uname -a
Linux kali 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05) x86_64 GNU/Linux
(kali㉿kali)-[~]
$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 39 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 2
On-line CPU(s) list: 0,1
Vendor ID: GenuineIntel
Model name: 12th Gen Intel(R) Core(TM) i5-1235U
CPU family: 6
Model: 154
Thread(s) per core: 1
Core(s) per socket: 2
Socket(s): 1
Stepping: 4
Microcode version: 0xffffffff
BogoMIPS: 4992.00
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp lm constant_tsc rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq

2.B


kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ free -h
total used free shared buff/cache availa
ble
Mem: 1.9Gi 921Mi 475Mi 11Mi 725Mi 1.
0Gi
Swap: 953Mi 0B 953Mi
(kali㉿kali)-[~]
$ df -h
Filesystem Size Used Avail Use% Mounted on
udev 889M 0 889M 0% /dev
tmpfs 197M 972K 196M 1% /run
/dev/sda1 79G 16G 59G 21% /
tmpfs 985M 4.0K 985M 1% /dev/shm
none 1.0M 0 1.0M 0% /run/credentials/systemd-journald.service
tmpfs 985M 64K 985M 1% /tmp
none 1.0M 0 1.0M 0% /run/credentials/getty@tty1.service
tmpfs 197M 108K 197M 1% /run/user/1000
(kali㉿kali)-[~]
$ uptime
12:48:05 up 24 min, 1 user, load average: 0.74, 0.56, 0.47
(kali㉿kali)-[~]
$

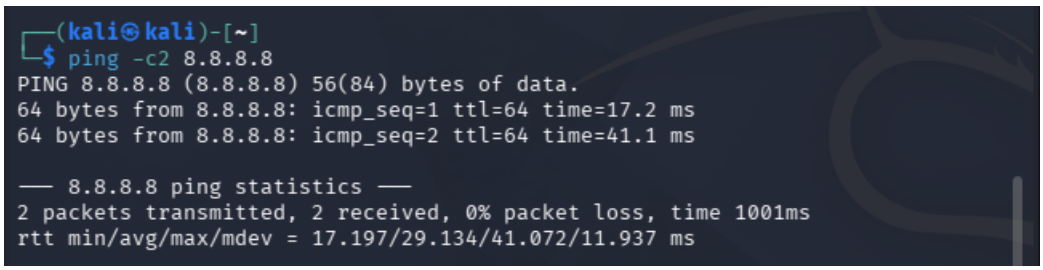
2.C Based on the terminal output:

- **Total RAM: 1.9 GiB** (about **2 GB**)
- **Main partition (**/dev/sda1**) disk space: 79 GB total**
- **Disk space available: 59 GB**
- **Disk space used: 16 GB** (21% used)

So, my VM has **~2 GB RAM** and **~59 GB of available disk space** on its main partition.

2.D I am using the Kernel**:** Linux 6.19.14 version.

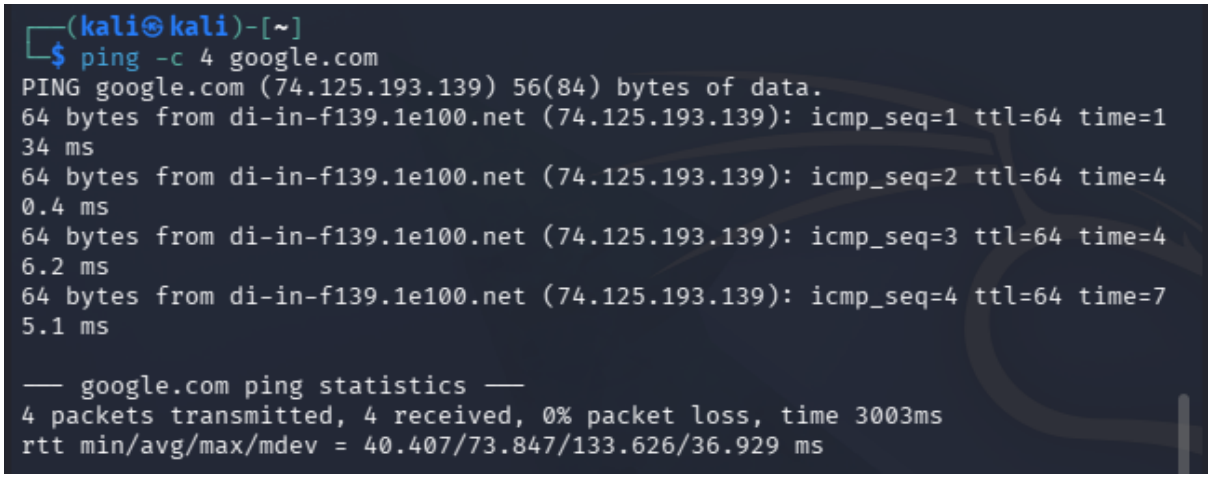
2.E



2.F

- **Average round-trip time: 17.293 ms**
- **Packet loss: 0%** — all **2 packets were received** successfully.

2.G



2.H

IP shown: 74.125.193.139 (resolved from google.com, hostname di-in-f139.1e100.net — that's Google's own reverse-DNS domain).

Round-trip time: This one's notably slower and jumpier than a typical 8.8.8.8 ping — min/avg/max here is 40.4 / 73.8 / 133.6 ms, with a large jitter (mdev) of ~37 ms. A ping to 8.8.8.8 from most locations usually comes back faster and much more consistent (often single digits to ~20-30 ms, with jitter of just a couple ms).

Why the difference is likely:

- **8.8.8.8 is anycast** — Google announces that address from many points of presence worldwide, so your packet gets routed to the *nearest* Google edge node topologically. It's built purely to be a fast, lightweight DNS resolver.
- **google.com resolves to a specific front-end web server IP** (via DNS, as you see here) — that IP may be serving from a different, possibly farther data centre, and it's handling full web-traffic infrastructure (load balancers, TLS termination, etc.) rather than being optimized as a bare low-latency anycast endpoint.
- The big variation between are 4 pings (134 → 40 → 46 → 75 ms) suggests the path is crossing more hops or shared/congested links, or requests are landing on different backend nodes each time — whereas 8.8.8.8's anycast routing tends to keep hitting the same nearby node consistently, giving much flatter, lower RTTs.

2.I

-c 4 is the count flag for ping — it tells ping to send exactly 4 ICMP echo requests and then stop.